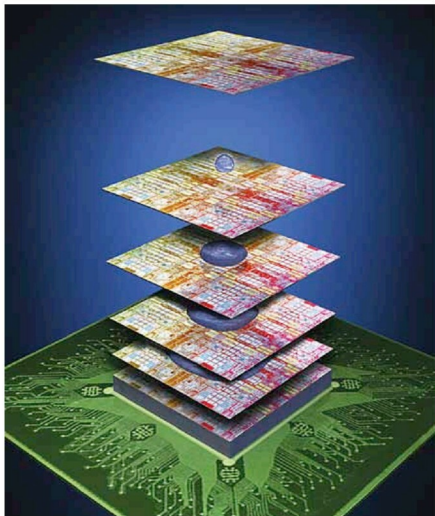


# Why 3D Chips are the Future of Electronics

Three-dimensional integrated circuits or 3D chips are believed to be the best way to keep Moore's Law ticking. These sound ideal for next-generation designs, provided manufacturing beefs up and costs go down. Here we try to answer some frequently-asked questions about 3D chips—and whether they really are 3D



3D IC stacking (Image courtesy: [www.i-micronews.com](http://www.i-micronews.com))

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On an average, a chip holds over one billion transistors today. Intel's 10-core Xeon Westmere-EX packs 2.6 billion tran-

sistors into a 512mm<sup>2</sup> area using 32nm technology. But even that is not enough. The thirst for more features and processing power is unquenchable, but how much more can you pack into a chip? If transistors get any

smaller, their reliability could become questionable. Plus, if you need to make chips denser, the fabrication costs might also become unreasonably high.

Such issues have led to the adoption of three-dimensional (3D) integrated circuit technology, wherein two or more dice are stacked atop each other and linked with suitable interconnects. You can draw a parallel with skyscrapers, which help overcome space shortage within large cities, and elevators that make it possible for people and even vehicles to easily travel to any of the floors.

## What are the main benefits of 3D chips?

1. 3D chips save space. Dice that used to be centimetres apart on a chip can now be placed millimetres apart, and that too vertically.
2. Since the dice are close by, the interconnect length is significantly smaller. This results in low latency and higher performance.
3. Stacking helps overcome the memory-to-processor performance bottleneck that plagues 2D ICs.
4. 3D interconnects permit data to be moved both horizontally and vertically. This helps boost performance by 30 to 40 per cent.
5. Bandwidth is much higher as it is possible to have a large number of vertical vias between the layers. Moreover, interconnects between dice in the 3D IC do not use input/output (I/O) buffers or bond wires.
6. 3D chips consume 40 to 50 per